# Math 121: Commutative Algebra

Prishita Dharampal

Worksheet 2.2: Algebraic Sets

**Theorem 0.1** (Hilbert's Nullstellensatz I). *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \ldots, x_n]$ is of the form $\langle x_1 - a_1, \ldots, x_n - a_n \rangle$ for some point $(a_1, \ldots, a_n) \in \mathbb{K}^n$.*

Note this is just the first of a series of theorems that will go by Hilbert's Nullstellensatz. Recall that a $\mathbb{K}$-algebra is a ring $R$ together with a specified ring homomorphism $\mathbb{K} \to R$. A morphism of $\mathbb{K}$-algebras $\varphi : R \to S$ is a ring homomorphism that preserves scalars, i.e. $\varphi(c \cdot 1_R) = c \cdot 1_S$ for every $c \in \mathbb{K}$. In particular, $\mathbb{K}[x_1, \ldots, x_n]$ is a $\mathbb{K}$-algebra, every evaluation map $\mathrm{ev}_p : \mathbb{K}[x_1, \ldots, x_n] \to \mathbb{K}$ is a $\mathbb{K}$-algebra homomorphism, and if $\mathfrak{m}$ is a maximal ideal then $R/\mathfrak{m}$ is naturally a field equipped with a $\mathbb{K}$-algebra structure. When we say below that a residue field is "isomorphic to $\mathbb{K}$," we mean *as a $\mathbb{K}$-algebra*.

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**Problem 1. Evaluation Maps.** For a point $p = (a_1, \ldots, a_n) \in \mathbb{K}^n$ the *evaluation-at-$p$* map is given by

$$\begin{array}{ccc} \mathbb{K}[x_1, \ldots, x_n] & \xrightarrow{\mathrm{ev}_p} & \mathbb{K} \\ f & \longmapsto & f(p) \end{array}.$$

(a) Prove that $\mathrm{ev}_p$ is a ring homomorphism and find generators for $\mathfrak{m}_p := \ker(\mathrm{ev}_p)$.

(b) Prove that there is a bijection, induced by these evaluation maps, between $\mathbb{K}^n$ and the subset of $\mathrm{Spec}(\mathbb{K}[x_1, \ldots, x_n])$ consisting of maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$-algebra.

(c) Find a maximal ideal of $\mathbb{R}[x, y]$ that is not of the form $\mathfrak{m}_p$ for any $p \in \mathbb{R}^2$.

(d) Restate Hilbert's Nullstellensatz I (above) in terms of $\mathfrak{m}_p$.

*Solution.*

(a) To show that $\mathrm{ev}_p$ is a ring homomorphism, we need to show that for all $f_1, f_2 \in \mathbb{K}[x_1, \cdots, x_n]$ the three conditions hold,

**Additivity:**

$$\mathrm{ev}_p(f_1 + f_2) = (f_1 + f_2)(p) = f_1(p) + f_2(p) = \mathrm{ev}_p(f_1) + \mathrm{ev}_p(f_2)$$

**Multiplicativity:**

$$\mathrm{ev}_p(f_1 \cdot f_2) = (f_1 \cdot f_2)(p) = f_1(p) \cdot f_2(p) = \mathrm{ev}_p(f_1) \cdot \mathrm{ev}_p(f_2)$$

And $\mathrm{ev}_p(1) = 1$ since $1$ is a constant function.

(b) **TODO:**

(c) **TODO:**

(d) **TODO:**

**Problem 2. Algebraic Sets.**
Let $\mathcal{S} \subset \mathbb{K}[x_1, \ldots, x_n]$ be a subset. Define the corresponding *algebraic set* in $\mathbb{K}^n$ to be:

$$\mathbb{V}(\mathcal{S}) := \left\{ p \in \mathbb{K}^n \;\middle|\; f(p) = 0 \text{ for all } f \in \mathcal{S} \right\}.$$

(a) Prove that $\mathbb{V}$ is order-reverse, i.e., if $\mathcal{S} \subset \mathcal{T}$ then $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$.

(b) Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$.

(c) Describe each of the following algebraic sets, and when possible draw a picture.

- (i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.
- (ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.
- (iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.
- (iv) $\mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.
- (v) $\mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.
- (vi) $\mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.
- (vii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x]$.
- (viii) $\mathcal{S} := \{x^2 + 1\} \subset \mathcal{C}[x]$.
- (ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \ldots, x_n]$.

(d) Explain why $\mathrm{SL}_n(\mathbb{K})$ has the structure of an algebraic set.

(e) Explain why the set of $m \times n$ matrices of rank $< t$ is an algebraic set contained in $\mathbb{K}^{m \times n}$.

*Solution.*

(a) Assume $\mathcal{S} \subset \mathcal{T}$. Let $p \in \mathbb{V}(\mathcal{T})$.

Then, for all $f \in \mathcal{T}$, $f(p) = 0$. But since $\mathcal{S} \subset \mathcal{T}$ this implies $g(p) = 0, \forall g \in \mathcal{S}$. Hence, $p \in \mathbb{V}(\mathcal{S})$. Since $p$ was an arbitrary point in $\mathbb{V}(\mathcal{T})$ we conclude that $\mathbb{V}$ is order reversing.

(b) By definition,

$$\mathcal{S} \subseteq \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}.$$

And from part (a) we have,

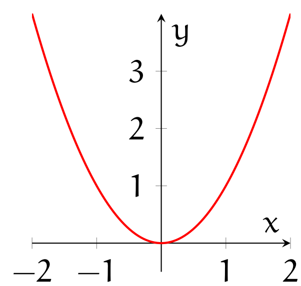
$$\mathbb{V}(\mathcal{S}) \supseteq \mathbb{V}(\langle \mathcal{S} \rangle) \supseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right).$$

To show $\mathbb{V}(\langle S \rangle) \subseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right)$ we let $p \in \mathbb{V}(\langle \mathcal{S} \rangle)$. That is for all $f \in \langle S \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$, $f(p) = 0$, i.e. $p \in \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right) \implies \mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \left(\sqrt{\langle \mathcal{S} \rangle}\right)$.

To show that $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\langle \mathcal{S} \rangle)$ we let $p \in \mathbb{V}(\mathcal{S})$. That is for all $f \in \mathcal{S}$, $f(p) = 0$ but since $\langle S \rangle$ is the set of linear combinations of functions that evalute to zero at $p$, they also evaluate to zero at $p$ and $p \in \mathbb{V}(\langle \mathcal{S} \rangle)$.

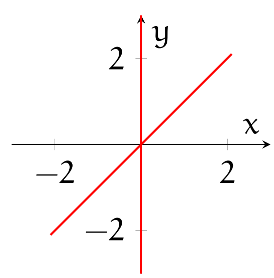
(c) (i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$




(ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

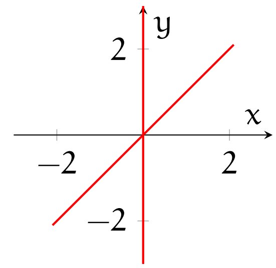
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0,\ x = y\}$$




(iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

Note that $x^6 - 3x^5y + 3x^4y^2 - x^3y^3 = x^3(x - y)^3$.

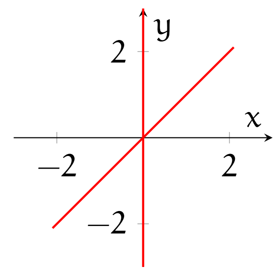
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0,\ x = y\}$$




(iv) $\mathcal{S} \coloneqq \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

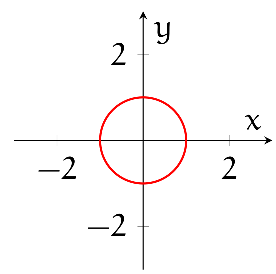
Note that $x^4 - 2x^3y + x^2y^2 = x^2(x - y)^2$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0,\ x = y\}$$




(v) $\mathcal{S} \coloneqq \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$




(vi) $\mathcal{S} \coloneqq \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.

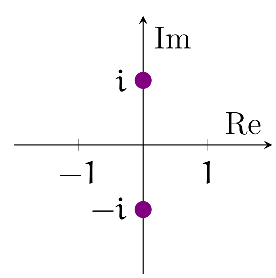
$$\mathbb{V}(\mathcal{S}) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z^2\}$$

(vii) $\mathcal{S} \coloneqq \{x^2 + 1\} \subset \mathbb{R}[x]$.

$\mathbb{V}(\mathcal{S}) = \emptyset$ since this equation has no solutions over $\mathbb{R}[x]$

(viii) $\mathcal{S} \coloneqq \{x^2 + 1\} \subset \mathcal{C}[x]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



(ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \ldots, x_n]$.
$\mathbb{V}(\mathcal{S}) = \ker(A)$, where $A$ is a $m \times n$ matrix where the $i$-th row is $(a_{i_1}, \cdots, a_{i_n})$

(d) $SL_n(\mathbb{K})$ is the set of $n \times n$ matrices with determinant $1$, and the determinant is a polynomial function on $\mathbb{K}^{n \times n}$. Therefore we can define the polynomial

$$f = \det(A) - 1 \in \mathbb{K}[x_1, \cdots, x_{n \times n}]$$

where $A$ is a $n \times n$ matrix, and $SL_n(\mathbb{K}) = \mathbb{V}(\{f\})$.

(e) **TODO:**

**Problem 3. Defining Ideal of an Algebraic Set.**
Let $V \subset \mathbb{K}^n$ be an algebraic set. The *defining ideal* of $V$ is the ideal in $\mathbb{K}[x_1, \ldots, x_n]$ given by

$$\mathbb{I}(V) \coloneqq \left\{ f \in \mathbb{K}[x_1, \ldots, x_n] \;\middle|\; f(p) = 0 \text{ for all } p \in V \right\}.$$

(a) Prove that $\mathbb{I}(V)$ is in fact an ideal.

(b) Prove that $\mathbb{I}(V)$ is radical.

(c) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, show that $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$.

(d) Let $\mathbb{K} = \mathbb{R}$. Give an example to show that the inclusion $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.

(e) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, show that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(V))$.

*Solution.*

(a) Let $g \in \mathbb{K}[x_1, \cdots, x_n]$ and $f_1, f_2 \in \mathbb{I}(V)$. That is, $f_1(p) = f_2(p) = 0$.

**Subgroup Under Addition:** Note that,

$$(f_1 + f_2)(p) = f_1(p) + f_2(p) = 0$$

The zero polynomial $0(p) = 0$ acts as the identity, and

$$f_1(p) + (-f_1)(p) = 0(p) = 0$$

gives us the additive inverse.

**Closed Under Multiplication:** Notice,

$$(g \cdot f)(p) = g(p) \cdot f_1(p) = g(p) \cdot 0 = 0$$

Thus $g \cdot f \in \mathbb{I}(V)$ and $\mathbb{I}(V)$ is an ideal.

(b) We want to show that $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

Note that the containment $\mathbb{I}(V) \subseteq \sqrt{\mathbb{I}(V)}$ is obvious. To show the opposite containment take $f \in \sqrt{\mathbb{I}(V)}$. Then for some $n$ we have,

$$f^n(p) = (f(p))^n = 0$$

But since $\mathbb{K}$ is a domain, this implies that $f(p) = 0$. Hence $f \in \mathbb{I}(V)$, $\mathbb{I}(V) \supseteq \sqrt{\mathbb{I}(V)}$, and $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

(c) Let $V = \mathbb{V}(\mathcal{S})$. We know that $\langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$. So showing that $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ suffices. Note that $\mathbb{I}(V) = \mathbb{I}(\mathbb{V}(\mathcal{S}))$ which is the set of all functions that evaluate to $0$ at $p$, for all $p \in \mathbb{V}(\mathcal{S})$. And $\mathbb{V}(\mathcal{S})$ is the set of all points $p$ such that all functions $f \in \mathcal{S}$ evaluate to $0$ at $p$. Let $f \in \sqrt{\langle \mathcal{S} \rangle}$. Then by definition for some $n$, $f^n in \S$. We can then write,

$$f^n = \sum_i h_i(p) g_i(p), \quad h_i \in \mathbb{K}[x_1, \cdots, x_n], g_i \in \mathcal{S}$$

But then,

$$f^n(p) = \sum_i h_i(p) g_i(p) = \sum_i h_i(p) \cdot 0 = 0$$

Moreover since $\mathbb{K}$ is a domain, $f^n(p) = 0 \implies f(p) = 0 \implies f$ is a function that evaluates to $0$ at $p$, hence $f \in \mathbb{I}(V)$.

(d) Consider the set $\mathcal{S} = \{x^2 + 1\}$. We have,

$$\sqrt{\langle \mathcal{S} \rangle} = \sqrt{\langle x^2 + 1 \rangle}$$

And, because the polynomial has no solutions over $\mathbb{R}$, we have $V = \mathbb{V}(\mathcal{S}) = \emptyset$. But $\mathbb{I}(\mathbb{V}(\mathcal{S})) = \mathbb{R}[x]$. Also note that since $\langle \mathcal{S} \rangle$ is a prime ideal ($x^2 + 1$ is irreducible), we have $\langle x^2 + 1 \rangle = \sqrt{\langle x^2 + 1 \rangle} \subset \mathbb{R}[x]$. Hence, the containment $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.

(e) Let $V = \mathbb{V}(\mathcal{S})$.

$\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$

This containment is obvious by construction. $\mathbb{I}(\mathbb{V}(\mathcal{S}))$ is the set of all functions that vanish at all points in $\mathbb{V}(\mathcal{S})$. And $\mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$ is the set of all points that all functions in $\mathbb{I}(\mathbb{V}(\mathcal{S}))$ vanish at. So, $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\mathbb{I}(V))$.

$\mathbb{V}(\mathcal{S}) \supseteq \mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$

Note that by definition $\mathcal{S} \subseteq \mathbb{I}(\mathbb{V}(\mathcal{S}))$. Then by 2(a) $\mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S}))) \subseteq \mathbb{V}(\mathcal{S})$.

Hence, $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$.

So far you have built two order-reversing operations:

$$\mathcal{S} \longmapsto \mathbb{V}(\mathcal{S}) \quad \text{and} \quad V \longmapsto \mathbb{I}(V).$$

You have shown that passing from equations to a zero set only remembers the radical of the ideal they generate, and Question 4d shows that over a non-algebraically closed field this can lose information. The next theorem says that over an algebraically closed field this is the only ambiguity.

**Theorem 0.2** (Hilbert's Nullstellensatz II)**.** *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}$.*

At this point the basic affine dictionary is taking shape; Question (4) asks you to make the correspondence between algebraic sets and radical ideals completely explicit. After that we turn to coordinate rings, which package an algebraic set $V$ into algebra and will let us recover from $\mathbb{K}[V]$ its points, its topology, and ultimately its morphisms.

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**Problem 4**. Let $\mathbb{K}$ be algebraically closed. Use Hilbert's Nullstellensatz II (above) to give an explicit inclusion-reversing bijection between algebraic sets in $\mathbb{K}^n$ and radical ideals.

*Solution.*

Let $\mathcal{S} \subset \mathbb{K}[x_1, \cdots, x_n]$.

$\mathbb{V}(\mathcal{S}) \xrightarrow{\mathbb{I}} \sqrt{\langle \mathcal{S} \rangle}$

$\mathbb{V}(\mathcal{S}) \xleftarrow{\mathbb{V}} \sqrt{\langle \mathcal{S} \rangle}$

**TODO:** Diagram.

**Problem 5. Coordinate Rings.**

Let $V \subset \mathbb{K}^n$ be an algebraic set and $\mathcal{R} = \mathrm{Hom}_{\mathrm{Set}}(V, \mathbb{K})$ be the ring of all (set) functions from $V$ to $\mathbb{K}$. In this question we study the restriction map from $\mathbb{K}[x_1, \dots, x_n]$ to $\mathcal{R}$ given by:

$$\begin{aligned} \mathbb{K}[x_1, \dots, x_n] &\xrightarrow{\ \rho\ } \mathcal{R} \\ f &\longmapsto f|_V \end{aligned}.$$

Here we are viewing $f \in \mathbb{K}[x_1, \dots, x_n]$ as defining a function $\mathbb{K}^n \to \mathbb{K}$ given by $p \mapsto f(p)$. We call the image of $\rho$ the *coordinate ring* of $V$ and denote it $\mathbb{K}[V]$.

(a) Prove that $\rho$ is in fact a ring homomorphism.

(b) Verify that each of the following functions lies in the coordinate ring of the given algebraic set.

  (i) Let $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{R}^2$ and consider the function $V \to \mathbb{R}$ given by $(x, y) \mapsto e^{x^2+y^2}$.

  (ii) Let $V = \mathbb{V}(xy - 1) \subset \mathbb{K}^2$ and consider the function $V \to \mathbb{K}$ given by $(x, y) \mapsto \frac{1}{x}$.

(c) Identify the kernel of $\rho$ and use it to give an explicit presentation for $\mathbb{K}[V]$ as a $\mathbb{K}$-algebra.

(d) The $i$-th coordinate of $\mathbb{K}^n$ is the linear functional $e_i^\vee : \mathbb{K}^n \to \mathbb{K}$ sending $(a_1, \dots, a_n) \mapsto a_i$. Explain why $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$, and under the presentation given in part (c), which polynomial corresponds to $e_i^\vee$.

(e) Explain why $\mathbb{K}[V]$ is called the coordinate ring of $V$. Hint: Think about the previous part.

(f) Describe the coordinate ring for the following algebraic sets:

  (i) $\mathbb{V}(y - x^2) \subset \mathbb{K}^2$.

  (ii) $\mathbb{V}(y^2, xy, x^{17}) \subset \mathbb{K}^2$.

  (iii) $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subset \mathbb{K}^2$.

  (iv) The set of $3 \times 3$ matrices over $\mathbb{F}_5$ of determinant $1$ and trace $0$. (Hint/Warning: There will likely be 9 more polynomials in the defining ideal than you expect, coming from the fact that finite fields have a special automorphism. This highlights another way we might get a counterexample for Question 4d.)

*Solution.*

(a) Let $f_1, f_2 \in \mathbb{K}[x_1, \cdots, x_n]$. Note that addition and multiplication is defined pointwise in $R$. To see if $\rho$ is a ring homomorphism, we check if the three properties hold:

**Addition:**

$$\rho(f_1) + \rho(f_2)(p) = f_1|_V(p) + f_2|_V(p) = (f_1 + f_2)|_V(p) = \rho(f_1 + f_2)(p)$$

**Multiplication:**

$$\rho(f_1) \cdot \rho(f_2)(p) = f_1|_V(p) \cdot f_2|_V(p) = (f_1 \cdot f_2)|_V(p) = \rho(f_1 \cdot f_2)(p)$$

**Unital:**

$$\rho(1)(p) = 1|_V(p) = 1$$

(b) (i) Let $f : (x, y) \mapsto e^{x^2+y^2}$. Note that on $V$, $x^2 + y^2 = 1$ then, $f|_V = e^{x^2+y^2} = e^1$. Which behaves like the constant polynomial function $g(x, y) = e$ and hence lies in the ring of coordinates of $V$.

(ii) Let $f : (x, y) \mapsto \frac{1}{x}$. Note that on $V$, $\frac{1}{x} = y$, and $x \neq 0$ because $\mathbb{K}$ is a domain. Then, $f|_V = \frac{1}{x} = y$. Which behaves like the polynomial function $g(x, y) = y$ and hence lies in the ring of coordinates of $V$.

(c) The kernel of $\rho$ is $\mathbb{I}(V)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x_1, \cdots, x_n]/\mathbb{I}(V)$$

(d) The coordinate ring of $\mathbb{K}^n$, $\mathbb{K}[\mathbb{K}^n]$ is the ring of all set functions $\mathbb{K}^n \to \mathbb{K}$.

$$e_i^{\vee}(x_1, \cdots, x_n) = x_i$$

which corresponds to the polynomial function $g(x_1, \cdots, x_n) = x_i$. Hence, $e_i^{\vee} \in \mathbb{K}[\mathbb{K}^n]$. Under the presentation given in part (c) we have,

$$\mathbb{K}[\mathbb{K}^n] \cong \mathbb{K}[x_1, \cdots, x_n]/\mathbb{I}(\mathbb{K}^n) = \mathbb{K}[x_1, \cdots, x_n]/\{0\} = \mathbb{K}[x_1, \cdots, x_n]$$

So, $e_i^{\vee}$ corresponds to the coset $[x_i] = x_i$.

(e) $\mathbb{K}[V]$ is the coordinate ring of $V$ because every polynomial is a combination of coordinate functions restricted to $V$. And $\mathbb{K}[V]$ is generated by these functions on $V$.

$$\mathbb{K}[V] = \mathbb{K}[e_1^{\vee}|_V, \cdots, e_n^{\vee}|_V]$$

(f) (i) Let $V = \mathbb{V}(y - x^2)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\mathbb{I}(y - x^2) = \mathbb{K}[x, x^2] = \mathbb{K}[x]$$

(ii) Let $V = \mathbb{V}(y^2, xy, x^{17})$.
Note that $\mathbb{I}(V) = \sqrt{\langle y^2, xy, x^{17}\rangle}$.
Since $\mathbb{K}$ is a field, if $(a, b) \in V$ then

$$a^{17} = 0 \implies a = 0$$

and

$$b^2 = 0 \implies b = 0$$

Then, $V = \{(0, 0)\}$. And $I(V) = I(\{(0, 0)\}) = \langle x, y\rangle$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\mathbb{I}(\mathbb{V}(y^2, xy, x^{17})) = \mathbb{K}[x, y]/\langle x, y\rangle = \mathbb{K}$$

(iii) Let $V = \mathbb{V}(x^6 - 3x^5y + 3x^4y^2 + x^3y^3) = \mathbb{V}(x^3(x - y)^3)$.
Then if $(a, b) \in V$, either $a^3 = 0 \implies a = 0$ or $(x - y)^3 = 0 \implies x = y$.
Then $V = \{(0, a), (a, a) \mid a \in \mathbb{K}\} = \mathbb{V}(x) \cup \mathbb{V}(x - y)$.

$$\mathbb{I}(V) = \mathbb{I}(\mathbb{V}(x) \cup \mathbb{V}(x - y)) = \mathbb{I}(\mathbb{V}(x)) \cap \mathbb{I}(\mathbb{V}(x - y)) = \langle x\rangle \cap \langle x - y\rangle = \langle x(x - y)\rangle$$

And

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\langle x(x - y)\rangle$$

(iv) Let $A = (x_{ij}) \in SL_3(\mathbb{F}_5)$, and $V = \mathbb{V}(\det(A) - 1, \operatorname{tr}(A))$. Note that the trace formula is $x_{11} + x_{22} + x_{33}$, and we use fermat's little theorem to enforce staying in $\mathbb{F}_5$. Then the coordinate ring is,

$$\mathbb{K}[V] = \mathbb{F}_5[x_{11}, \cdots, x_{33}]/\langle \det(A) - 1, x_{11} + x_{22} + x_{33}, x_{ij}^5 - x_{ij}\rangle$$

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You should now think of $\mathbb{K}[V]$ as the algebraic avatar of $V$: its elements are precisely the polynomial functions on $V$, and the coordinate functions on $\mathbb{K}^n$ descend to generators of $\mathbb{K}[V]$. The next step is to recognize points of $V$ purely algebraically, either as $\mathbb{K}$-algebra maps $\mathbb{K}[V] \to \mathbb{K}$ or as maximal ideals whose residue fields are isomorphic to $\mathbb{K}$ as $\mathbb{K}$-algebras.

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**Problem 6**. Let $R$ be a finitely generated $\mathbb{K}$-algebra. Fix a presentation $R \cong \mathbb{K}[x_1, \ldots, x_n]/I$. Describe a natural bijection between the following three sets:

(i) points in $\mathbb{V}(I) \subset \mathbb{K}^n$,

(ii) maximal ideals in $R$ with residue field isomorphic to $\mathbb{K}$ as a $\mathbb{K}$-algebra, and

(iii) elements of $\mathrm{Hom}_{\mathbb{K}-\mathbf{alg}}(R, \mathbb{K})$, the set of morphisms from $R$ to $\mathbb{K}$ in the category of $\mathbb{K}$-algebras.

*Solution.*

Let $\mathbb{K}$ be algebraically closed.

**Claim:** Let $\pi : \mathbb{K}[x_1, \cdots, x_n] \to \mathbb{K}[x_1, \cdots, x_n]/I = R$, and let $J \subset R$ be maximal. We claim that $\pi^{-1}(J)$ is maximal in $\mathbb{K}[x_1, \cdots, x_n]$.

*Proof.* We know that $\pi^{-1}(J)$ is an ideal. Let $M$ be a maximal ideal containing $\pi^{-1}(J)$. Since $\pi$ is surjective, we know that $\pi(M)$ is an ideal (worksheet 1.1, Q4.4), and $J \subseteq \pi(M)$. Then, since $J$ is maximal, either $\pi(M) = J$ or $\pi(M)$ is the whole ring. $\pi(M) \neq R$ since ring homomorphisms are unital and $1 \notin M$. Hence, $M = \pi^{-1}(J)$ or $\pi^{-1}(J)$ is maximal. $\square$

$(i) \iff (ii)$

Recall that $\mathbb{V}(I)$ contains points $p = (a_1, \cdots, a_n)$ such that $f(p) = 0$ for all $f \in I$. Also,

$$\mathrm{ev}_p : \mathbb{K}^n \to \mathbb{K}$$

By Hilbert's Nullstellensatz I, we know that maximal ideals of $\mathbb{K}[x_1, \cdots, x_n]$ are of the form $\langle x_1 - a_1, \cdots, x_n - a_n \rangle$, for some point $(a_1, \cdots, a_n)$.

**Problem 7**. **Zariski Topology on Algebraic Sets.** Let $V \subseteq \mathbb{K}^n$ be an algebraic set. Define the *Zariski topology* on $V$ by declaring the closed sets to be the subsets of $V$ of the form $\mathbb{V}(I)$ for some ideal $I \subset \mathbb{K}[x_1, \ldots, x_n]$. In other words, the closed sets are precisely the *algebraic subsets* of $V$.

(a) Verify that this definition indeed determines a topology on $V$.

(b) For $\mathsf{f} \in \mathbb{K}[V]$, define
$$D(f) = \left\{ p \in V \;\middle|\; f(p) \neq 0 \right\}.$$
Show that $\{D(f)\}$ form a basis for the Zariski topology on $V$.

(c) For each point $p \in V$, let
$$\mathfrak{m}_p = \left\{ f \in \mathbb{K}[V] \;\middle|\; f(p) = 0 \right\}$$
be the corresponding maximal ideal. Show that the map
$$\begin{aligned} V &\longrightarrow \operatorname{Spec}(\mathbb{K}[V]) \\ p &\longmapsto \mathfrak{m}_p \end{aligned}$$
is a homeomorphism onto its image, which consists of those maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$-algebra.

(d) If $\mathbb{K}$ is algebraically closed how can the preceding part be rephrased in terms of $\operatorname{mSpec}(\mathbb{K}[V])$?

(e) Is the Zariski topology on $V$ Hausdorff in general? If not, when is it?

*Solution.*

(a) $V = \mathbb{V}(I)$ for some $I \subset \mathbb{K}[x_1, \cdots, x_n]$. The closed sets of $V$ are of the form $\mathbb{V}(J)$, where $I \subset J \subset \mathbb{K}[x_1, \cdots, x_n]$. We need to check the following:

   (i) $\emptyset$ and $V$ are closed.
   Note $\mathbb{V}(I) = V$, and $\mathbb{V}(\mathbb{K}[x_1, \cdots, x_n]) = \emptyset$ is closed.

   (ii) Finite union of closed sets is closed.
   Let $\mathbb{V}(J), \mathbb{V}(K) \subseteq V$ be subsets. Then, $\mathbb{V}(I) \cup \mathbb{V}(J)$ is the set of points in $\mathbb{K}^n$ such that for all $p \in \mathbb{V}(I) \cup \mathbb{V}(J)$ either $f(p) = 0$ or $g(p) = 0$.

   (iii) Arbitrary intersection of closed sets is closed.

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By now you have recovered from $\mathbb{K}[V]$ not just the points of $V$, but also its Zariski topology. The next goal is to understand maps in the same way: a morphism of algebraic sets should be encoded contravariantly by a $\mathbb{K}$-algebra homomorphism of coordinate rings.

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**Problem 8. Morphisms of Algebraic Sets.** A *polynomial map* from $\mathbb{K}^n$ to $\mathbb{K}^m$ is a function of the form:

$$\Phi : \mathbb{K}^n \longrightarrow \mathbb{K}^m \quad p \mapsto (f_1(p), \ldots, f_m(p)).$$

where $f_1, \ldots, f_m \in \mathbb{K}[x_1, \ldots, x_n]$. A *morphism of algebraic sets* $V \to W$ is defined to be the restriction of such a polynomial map $\Phi$ with $\Phi(V) \subset W$. Fix a polynomial map $\Phi$ as above.

(a) Show that there is an induced $\mathbb{K}$-algebra homomorphism

$$\phi : \mathbb{K}[\mathbb{K}^m] \longrightarrow \mathbb{K}[\mathbb{K}^n] \quad \text{given by} \quad g \mapsto g \circ \Phi.$$

Describe $\phi$ explicitly in terms of the coordinate function generators of $\mathbb{K}[\mathbb{K}^m]$.

(b) Let $V \subset \mathbb{K}^n$ and $W \subset \mathbb{K}^m$ be algebraic sets. Show that $\Phi(V) \subset W$ if and only if $\phi(\mathbb{I}(W)) \subset \mathbb{I}(V)$.

(c) Deduce that $\Phi$ restricts to a morphism of algebraic sets $\Phi|_V : V \longrightarrow W$ if and only if the composition

$$\mathbb{K}[\mathbb{K}^m] \xrightarrow{\ \phi\ } \mathbb{K}[\mathbb{K}^n] \xrightarrow{\ \rho\ } \mathbb{K}[V]$$

factors through $\mathbb{K}[W]$. Here $\rho$ is the restriction map.

(d) For each of the given maps of coordinate rings, describe the corresponding map of algebraic sets. When possible draw pictures. Compare to the corresponding maps on spectra.

(i) $\mathcal{C}[x, y] \to \mathcal{C}[t]$ given by $x \mapsto t$ and $y \mapsto t$.

(ii) $\mathcal{C}[t] \to \mathcal{C}[x, y]$ given by $t \mapsto x$.

(iii) $\mathcal{C}[t] \to \mathcal{C}[x, y, t]/\langle xy - t\rangle$ given by $t \mapsto t$.

**Problem 9.** Prove that a morphism of algebraic sets is continuous with respect to the Zariski topology.

**Problem 10.** Show that the image of $\mathbb{K} \to \mathbb{K}^3$ sending $t \to (t, t^2, t^3)$ is an algebraic set defined by two polynomials. Verify your answer by constructing the corresponding ring map in *Macaualy2*.

**Problem 11**. Prove or disprove: The image of every polynomial map $\mathbb{K}^n \to \mathbb{K}^m$ is an algebraic set. (Warning: this might be field dependent. As a hint, if $\mathbb{K}$ is finite, which subsets of $\mathbb{K}^n$ are algebraic sets?)

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You now have the main algebra-geometry dictionary in hand. Over any field $\mathbb{K}$, every algebraic set $V \subset \mathbb{K}^n$ has a coordinate ring $\mathbb{K}[V]$, points of $V$ are the same as $\mathbb{K}$-algebra maps $\mathbb{K}[V] \to \mathbb{K}$, and morphisms $V \to W$ correspond contravariantly to $\mathbb{K}$-algebra homomorphisms $\mathbb{K}[W] \to \mathbb{K}[V]$. Thus the assignment

$$V \longmapsto \mathbb{K}[V]$$

always gives a contravariant bridge from geometry to algebra. When $\mathbb{K}$ is algebraically closed, Hilbert's Nullstellensatz says that every finitely generated reduced $\mathbb{K}$-algebra comes from an algebraic set, so this bridge becomes an anti-equivalence of categories.

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**Problem 12**. **Putting It All Together.** Assume that $\mathbb{K}$ is algebraically closed. Use the previous parts together with Hilbert's Nullstellensatz II to show that every finitely generated reduced $\mathbb{K}$-algebra is isomorphic to $\mathbb{K}[V]$ for some algebraic set $V$. Conclude that the assignment $V \mapsto \mathbb{K}[V]$ defines a (anti-) equivalence from the category of algebraic sets over $\mathbb{K}$ to the category of finitely generated reduced $\mathbb{K}$-algebras.